Speech Signal Processing: BSAI - 7th Semester

Lab 3 Task Report

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October 02, 2025

Report

Task 1: Z-transform

The Z-transform converts a discrete-time signal $x[n]$ into its frequency-domain representation:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

Part (a): For $x[n] = 1$:

$$X(z) = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = \frac{1}{1 - z^{-1}}, \quad |z| > 1.$$

Part (b): For $x[n] = n^4$:

$$X(z) = \sum_{n=0}^{\infty} n^4 z^{-n} = \frac{z(z^3 + 11z^2 + 11z + 1)}{(z-1)^5}, \quad |z| > 1.$$

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```
%%Z-transform
syms z n
a = ztrans(1/1^n)
f = n^4;
ztrans(f)
```

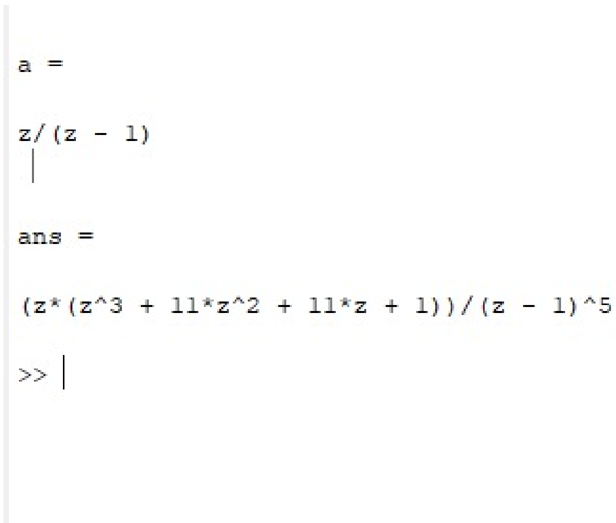

Figure 1: Z-transform outputs in MATLAB

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Task 2: Inverse Z-transform

The inverse Z-transform recovers the discrete-time signal from $X(z)$.

Part (a):

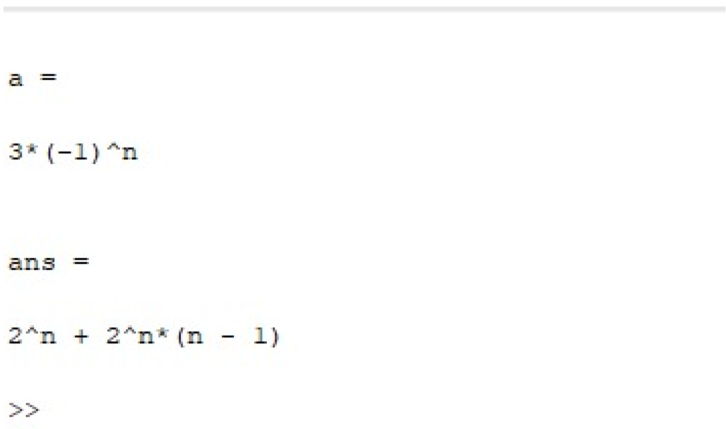
$$X(z) = \frac{3z}{z+1} \quad \Rightarrow \quad x[n] = 3(-1)^{n-1}u[n-1].$$

Part (b):

$$X(z) = \frac{2z}{(z-2)^2} \quad \Rightarrow \quad x[n] = 2(n+1)2^n u[n].$$

black

```
%%Inverse Z-transorm
syms Z n
a = iztrans(3*Z/(Z+1))
f = 2*Z/(Z-2)^2;
iztrans(f)
```



```
a =

3*(-1)^n


ans =

2^n + 2^n*(n - 1)

>>
```

Figure 2: Inverse Z-transform results in MATLAB

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Task 3: Pole-Zero Plot using zplane

We analyze:

$$H(z) = \frac{z^2 + z}{z^2 - 2z + 3}.$$

- Zeros: roots of numerator $\Rightarrow z = 0, -1$. - Poles: roots of denominator $\Rightarrow z = 1 \pm j\sqrt{2}$.

black

```
%%zplane
b = [0 1 1];
a = [1 -2 3];
roots(a)
roots(b)
zplane(b, a)
```

Analysis: Poles are located outside the unit circle, implying the system is **unstable**.

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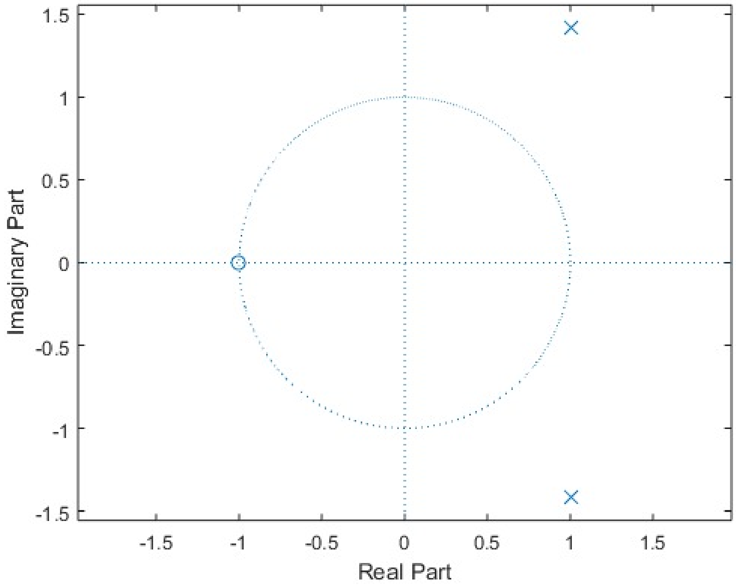

Figure 3: Pole-Zero plot of the system

Task 4: Frequency Response using freqz

The `freqz` function evaluates:

$$H(e^{j\omega}) = \frac{b(e^{j\omega})}{a(e^{j\omega})}.$$

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```
%%freqz
b = [0 1 1];
a = [1 -2 3];
freqz(b, a)
```

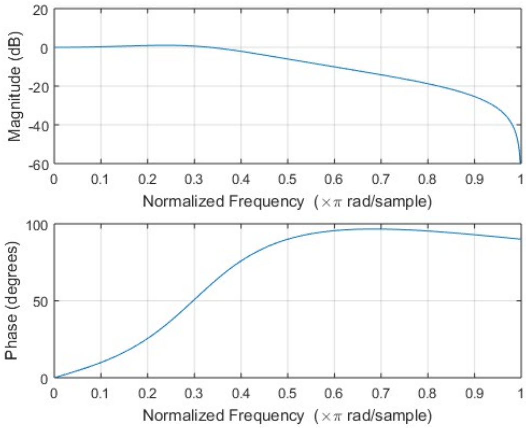

Figure 4: Magnitude and phase response using freqz

Analysis: The magnitude response shows resonance near the pole locations, while the phase response exhibits rapid changes around frequencies close to poles.

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Task 5: Impulse Response using impz

Impulse response $h[n]$ describes how the system reacts to $\delta[n]$.

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```
%%impz
b = [0 1 1];
a = [1 -2 3];
impz(b, a)
```

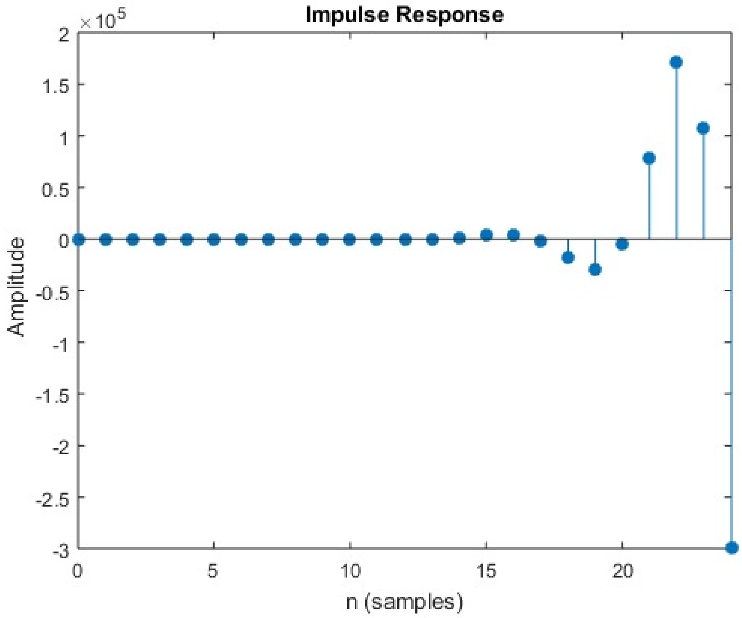

Figure 5: Impulse response of the system

Analysis: Since poles are outside the unit circle, the impulse response grows unbounded, confirming instability.

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Task 6: Pole-Zero and ROC Analysis

We analyze:

$$G(z) = \frac{2z^4 + 16z^3 + 44z^2 + 56z + 32}{3z^4 + 3z^3 - 15z^2 + 18z - 12}.$$

Using MATLAB: - Zeros and poles were computed. - System expressed in Second-Order Sections (SOS) form. - Region of Convergence (ROC) depends on poles: - For **causal system:** $|z| > \max |p_i|$. - For **anti-causal system:** $|z| < \min |p_i|$.

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```
%%Lab Task
num = [2 16 44 56 32];
den = [3 3 -15 18 -12];

[z, p, k] = tf2zp(num, den);
sos = zp2sos(z, p, k);

zplane(num, den);
```

```
maxPole = max(abs(p));
minPole = min(abs(p));

fprintf('ROC causal: |z| > %.4f\n', maxPole);
fprintf('ROC anti-causal: |z| < %.4f\n', minPole);
```

```
Zeros:
  -4.0000 + 0.0000i
  -2.0000 + 0.0000i
  -1.0000 + 1.0000i
  -1.0000 - 1.0000i

Poles:
  -3.2361 + 0.0000i
   1.2361 + 0.0000i
   0.5000 + 0.8660i
   0.5000 - 0.8660i

Gain:
    0.6667

Factored Form (SOS):
    0.6667    4.0000    5.3333    1.0000    2.0000   -4.0000
    1.0000    2.0000    2.0000    1.0000   -1.0000    1.0000

For a casual system, ROC: |z| > 3.2361
For a anti-casual system, ROC: |z| < 1.0000
```

Figure 6: Pole-Zero Plot of $G(z)$

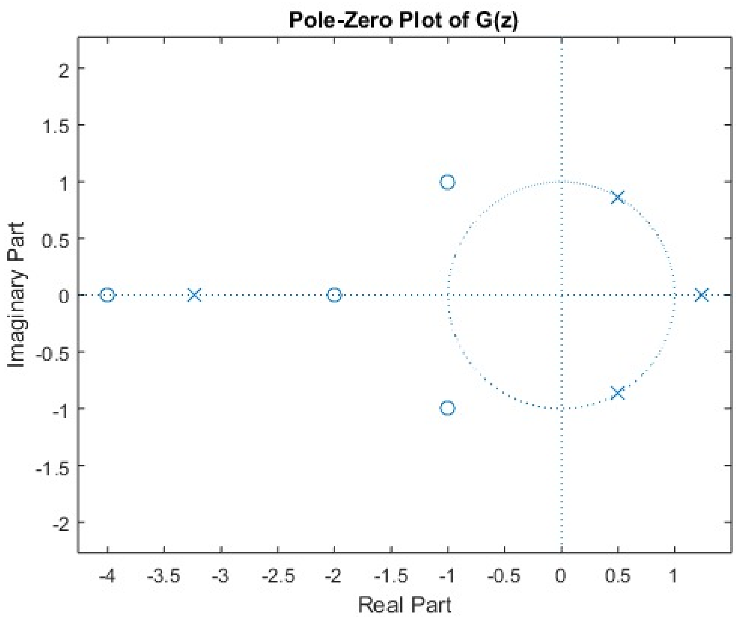

Figure 7: ROC analysis results

Analysis: - Poles closer to unit circle influence system behavior. - ROC is chosen based on causality. - If ROC includes unit circle, system is **stable**.

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Conclusion

In this lab, we applied Z-transform, inverse Z-transform, pole-zero analysis, frequency response, impulse response, and ROC analysis. The tasks highlighted the importance of Z-domain techniques for analyzing digital systems. MATLAB verified theoretical derivations, and the plots confirmed key properties such as stability, resonance, and system behavior.